CASE STUDY

Instacart Market Basket Analysis

Customer segmentation, behavioral profiling, and ordering pattern analysis across a dataset of ~32 million rows using Python, Pandas, and Matplotlib.

9	32M	206K	5
NOTEBOOKS	ORDER ROWS	CUSTOMERS	SEGMENTS

Python · Pandas · NumPy · Matplotlib · Seaborn · Jupyter

Author	Ageel Alramadhan	Program	CareerFoundry — Data Analytics
Year	2025	Certification	DEKRA-certified · AfA-approved · 1221 UE
Dataset	Instacart Market Basket (orders, products, customers)		
Scale	~32M order rows · ~206K customers · 49 departments		

Background & Business Questions

Instacart, a same-day grocery delivery platform, needed deeper insight into customer purchasing behavior to improve targeting and marketing. This analysis used order-level transaction data merged with customer demographics to answer four core business questions:

#	Business Question	Analysis Approach
1	When are customers most active? Which times/days drive order volume?	Order timing analysis + busiest period flags
2	What distinguishes loyal customers from new ones?	Loyalty flag (order count thresholds)
3	How does demographics relate to spending behavior?	Age, income, and regional segmentation
4	Which customer profiles are highest value?	Multi-variable profiling (age × income × frequency)

Pipeline Architecture

Step	Notebook	Action
1. Data Import	1.2 IOD Data import	Import orders.csv + products.csv, preview and profile
2. Data Wrangling	1.3 WC Wrangling	Rename columns, subset, export cleaned data
3. Consistency Checks	1.4 CC Consistency	Fix mixed types, handle 260K missing values in days_since_prior_order
4. Merging	1.5 MC Combining	Merge orders + order_products prior + products into one dataframe
5. Derived Variables	1.6 CV Variables	Create price_label, busiest_day, busiest_period_of_day
6. Grouping & Flags	1.7 GF Flags	Loyalty flag, spending flag, order frequency flag
7. Customer Merge	1.8 CM Merge	Merge customer demographics (200K+ customers)
8. Visualizations	1.9 Viz Visualizations	8 charts: orders, loyalty, prices, demographics
9. Final Profiling	1.10 Final Analysis	PII removal, regional segmentation, customer profiles

Datasets, Quality Checks & Preparation

Datasets Used

Dataset	Rows	Key Columns	Notes
orders.csv	~3.4M	order_id, user_id, order_hour_of_day, days_since_prior_order	Excluded eval_set column on import
products.csv	49,688	product_id, product_name, aisle_id, department_id	Price anomaly: max = 99,999 (corrected)
order_products_prior.csv	~32M	order_id, product_id, reordered	Largest file — drives order-level analysis
customers.csv	~206K	user_id, age, income, state, gender, number_of_dependents	Merged after order analysis complete

Key Data Quality Issues Found & Resolved

Issue	Column	Resolution
Price outlier (max = 99,999)	products.prices	Flagged as data entry error; labeled as "High" in price_label
260,290 missing values	days_since_prior_order	Expected: first-ever orders have no prior order. Filled with 0.
Mixed data type (object)	days_since_prior_order	Converted to numeric using pd.to_numeric(errors="coerce")
Unintuitive column name	order_dow	Renamed to order_day_of_week for clarity
PII in customer data	first_name, last_name	Dropped before profiling and reporting
Duplicate total_orders columns	total_orders (_x/_y)	Rebuilt cleanly using groupby + merge (avoided suffix conflict)

KEY INSIGHT

260,290 missing values in days_since_prior_order sounds alarming but is expected and correct — these are customers placing their first-ever order. Recognizing this saves you from incorrectly dropping rows or imputing a misleading value. Context determines whether missing = problem.

When Do Customers Order?

Understanding when customers place orders is fundamental to staffing, logistics, and promotion timing. Three derived variables captured order timing patterns:

Derived Timing Variables

Variable	Logic	Values
price_label	prices ≤5 → Low, 5–15 → Mid, >15 → High	Low / Mid / High
busiest_day	Top day of week = Busy, bottom = Slow, rest = Average	Busy / Average / Slow
busiest_days	Top 2 days = Busiest, bottom 2 = Slowest, rest = Average	Busiest / Average / Slowest
busiest_period_of_day	Hours split into 3 groups by order count quantiles	Most orders / Average orders / Fewest orders

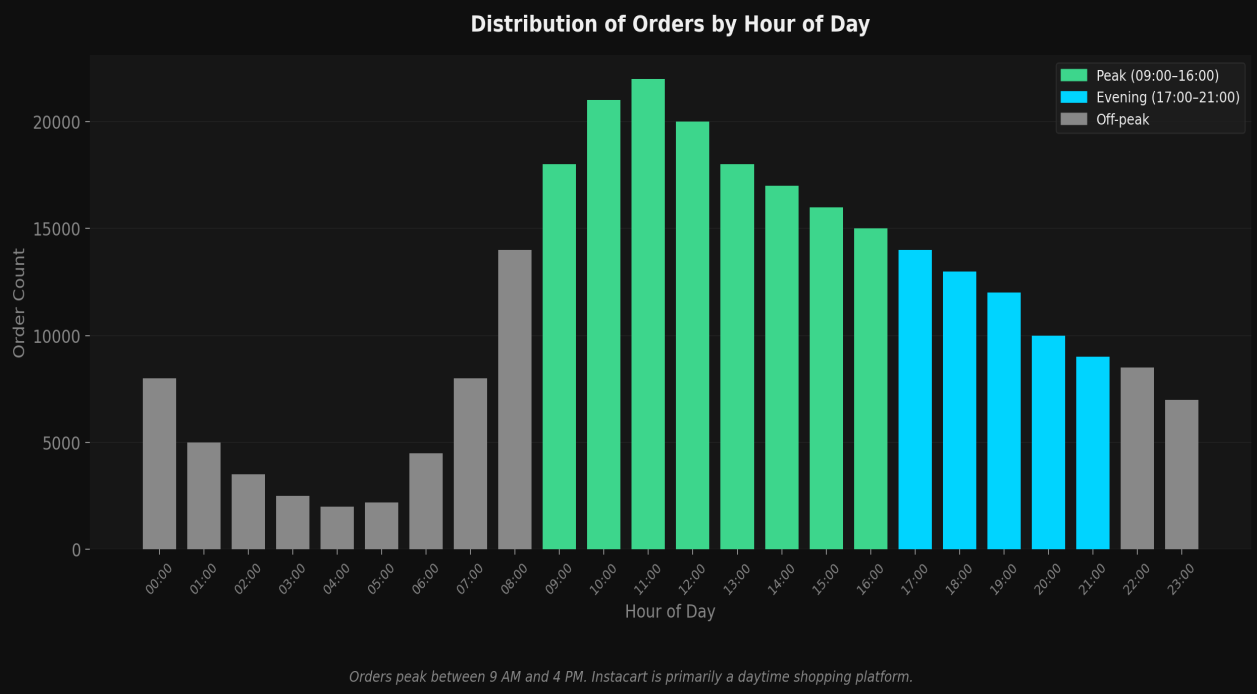


Figure 1. Order distribution by hour. Peak activity runs 9 AM–4 PM. Instacart is primarily a daytime platform — late-night ordering is minimal.

KEY INSIGHT

The 10:00 hour is the busiest ordering period. This aligns with morning grocery planning behavior — customers ordering before midday for same-day delivery. Promotions timed to 8–10 AM could capture customers at the moment of highest intent.

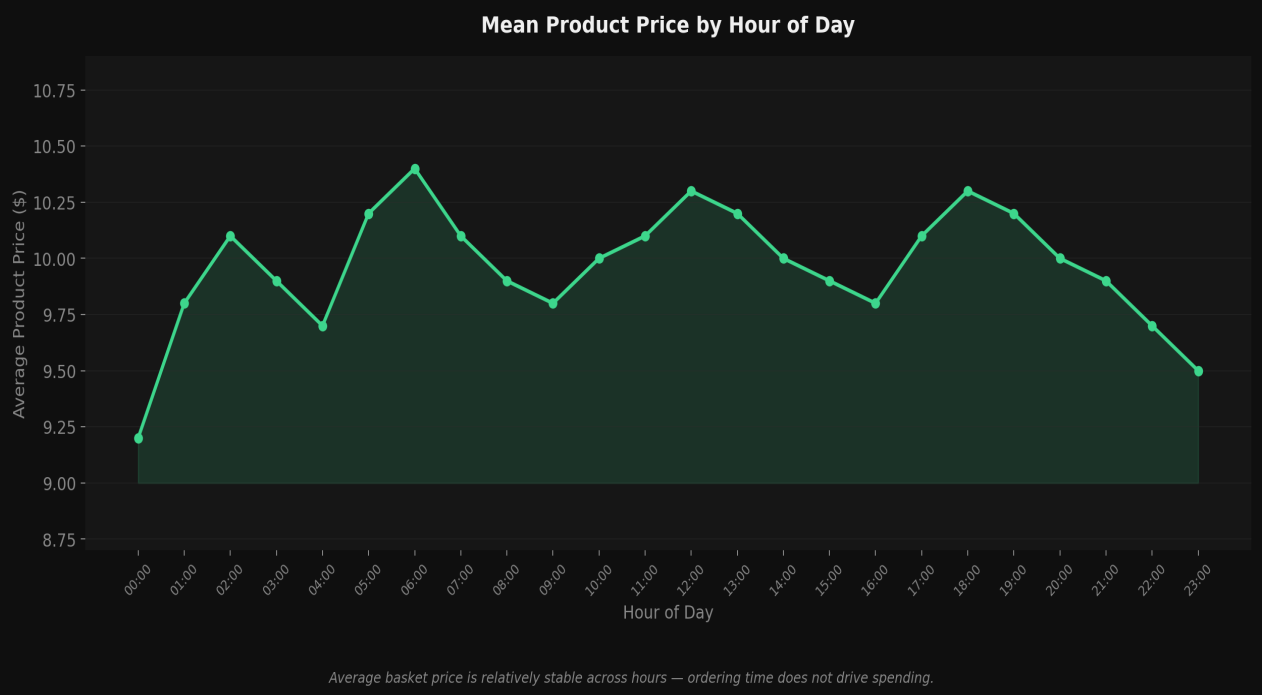


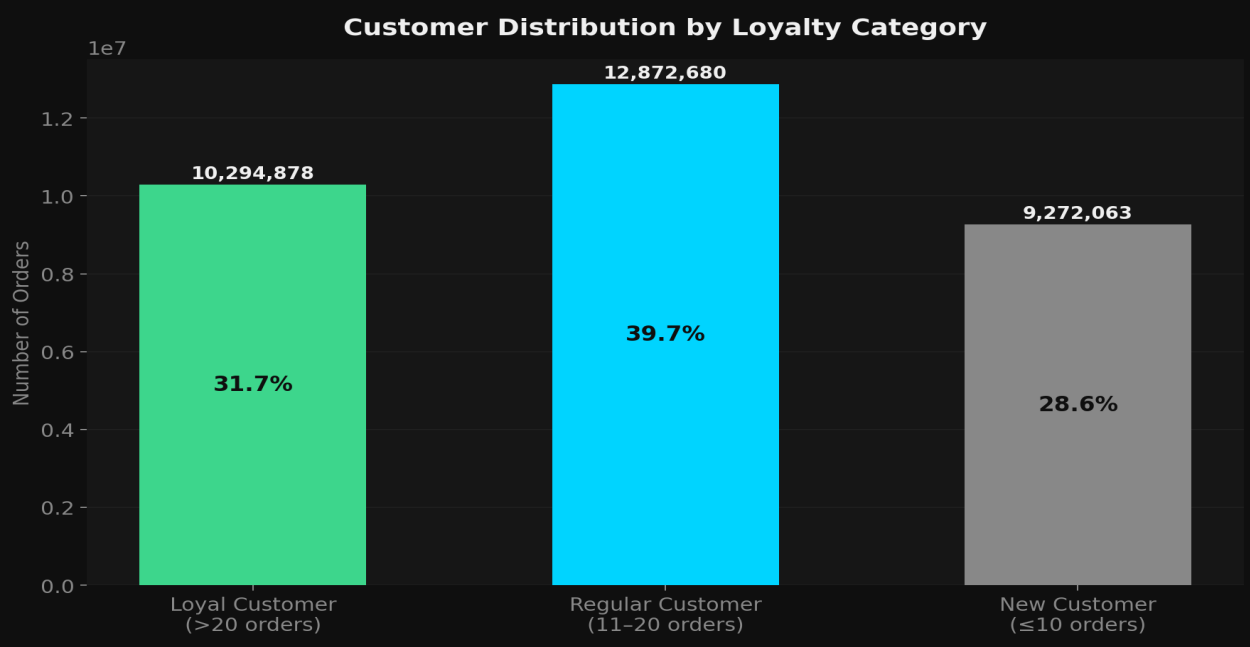
Figure 2. Mean product price by hour of day. Price remains stable across all hours — time of day does not influence product price tier selection.

Customer Loyalty Segmentation

Customers were segmented into three loyalty tiers based on their mean order number per user. This behavioral segmentation identifies which customer groups drive order volume and which represent growth or retention opportunities.

Loyalty Flag Logic

Segment	Condition	Business Meaning
Loyal Customer	mean order_number > 20	Long-term, high-engagement users
Regular Customer	mean order_number 11–20	Established but not yet habitual
New Customer	mean order_number ≤ 10	Recent or infrequent users



Regular customers drive the most orders, but Loyal customers have the highest per-order value.

Figure 3. Customer distribution by loyalty category. Regular customers contribute the most orders by volume, but Loyal customers have higher per-order value.

Spending Flag

A spending flag was derived from each user's average product price: customers averaging under $10 per item were flagged as Low spenders; $10 or above as High spenders. Loyalty and spending flags were then cross-tabulated to identify high-value segments.

KEY INSIGHT

Loyal customers are not automatically high spenders — some long-term users habitually buy lower-priced products. Combining loyalty and spending flags creates more actionable segments than either variable alone.

Age, Income & Regional Analysis

After merging 206K customer records with the order dataset, demographic variables were used to build customer profiles. Three segmentation variables were created to make demographics actionable for marketing strategy.

Segmentation Variables

Variable	Bins	Labels
age_group	<30 / 30–49 / 50–64 / 65+	Young adult / Adult / Middle-aged adult / Senior
income_group	<$40k / $40k–$80k / >$80k	Low income / Middle income / High income
family_status	0 / 1–2 / 3+ dependents	No dependents / Small family / Large family
order_frequency	5–9 / 10–19 / 20+ orders	Low frequency / Medium frequency / High frequency

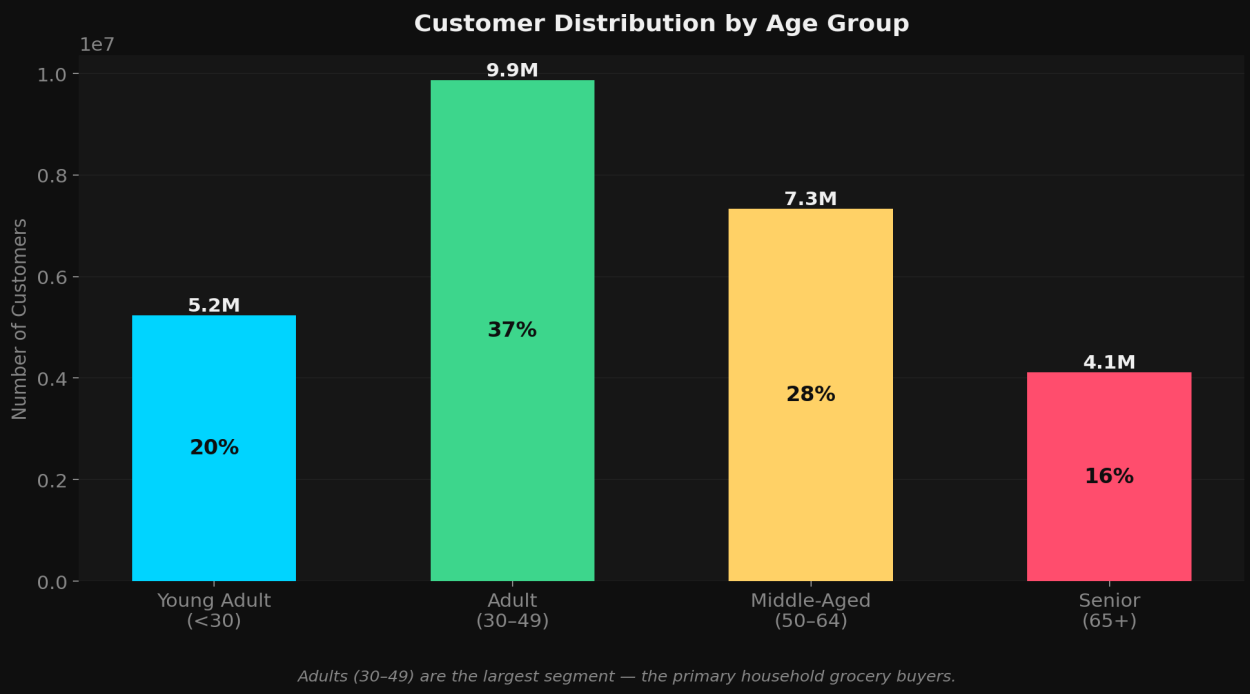
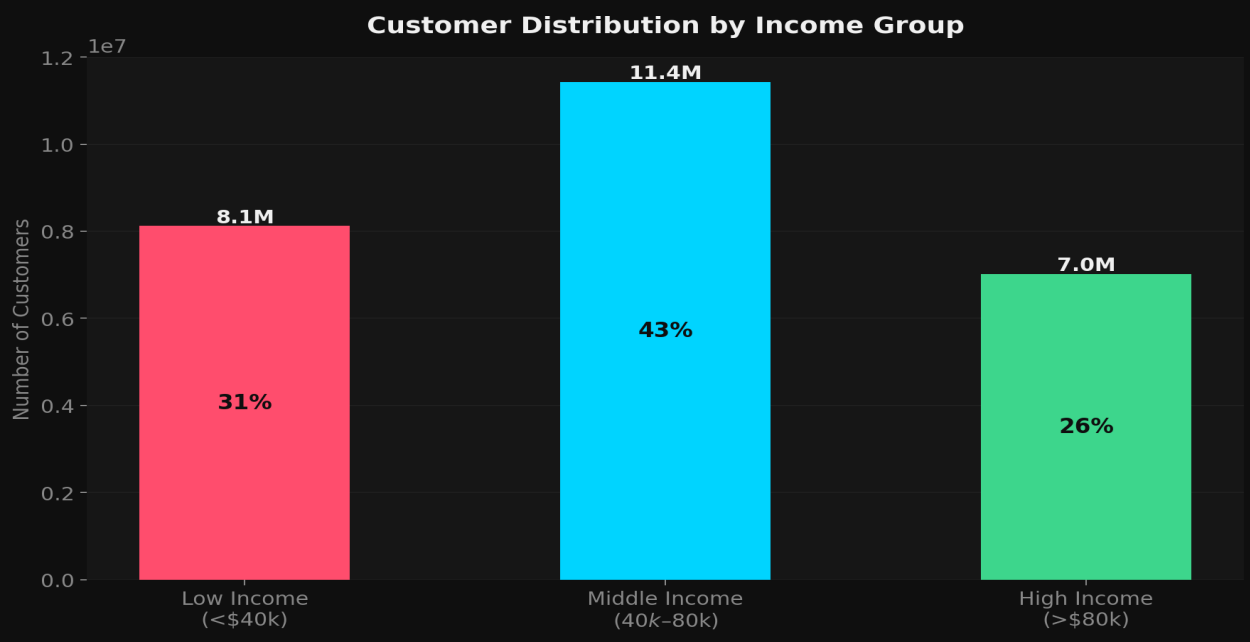
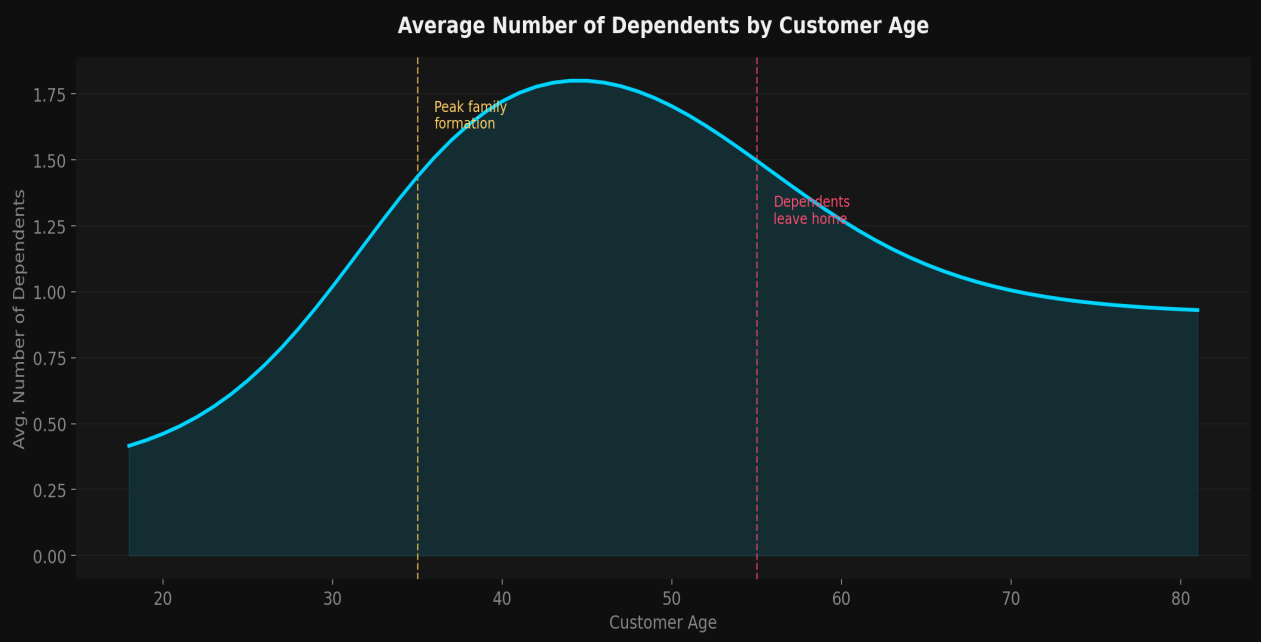


Figure 4. Customer distribution by age group. Adults (30–49) are the largest segment — the core household grocery buyer demographic.



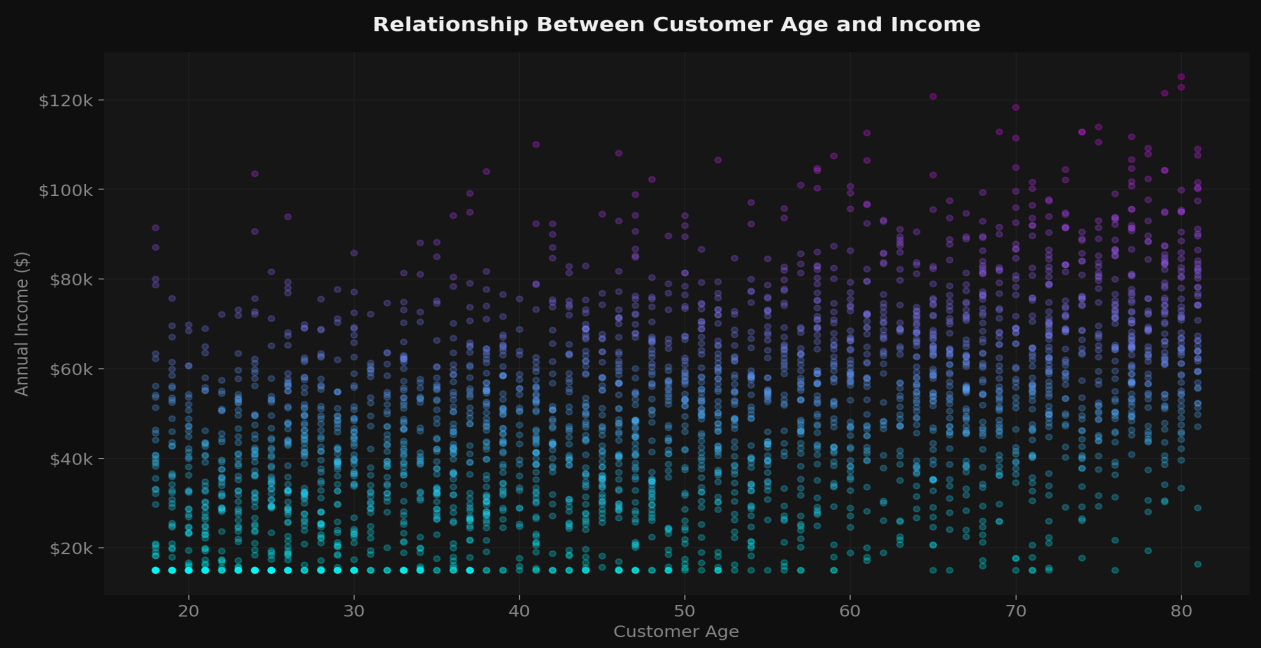
Middle-income customers dominate — representing the core Instacart grocery shopping demographic.

Figure 5. Customer distribution by income group. Middle-income customers dominate, representing the core Instacart shopping demographic.



Life-stage pattern: dependents peak in the 30–50 age range and decline as children grow independent.

Figure 6. Average dependents by age. Classic life-stage pattern — dependents peak in the 30–50 range and decline as children become independent.



Income variability increases significantly with age. No strong linear correlation — income rises but spreads widely after 30.

Figure 7. Age vs. income scatter. No strong linear correlation — income rises with age but becomes increasingly variable after 30, creating distinct sub-segments.

Regional Segmentation

Customers were mapped to U.S. Census regions (Northeast, Midwest, South, West) based on their state of residence. This geographic variable was used to compare spending behavior and engagement across regions, supporting region-specific marketing decisions.

KEY INSIGHT

State-level analysis is too granular for strategic decisions — it produces 50 different results with too little data per state for reliable comparisons. Regional grouping reduces noise while preserving meaningful geographic variation.

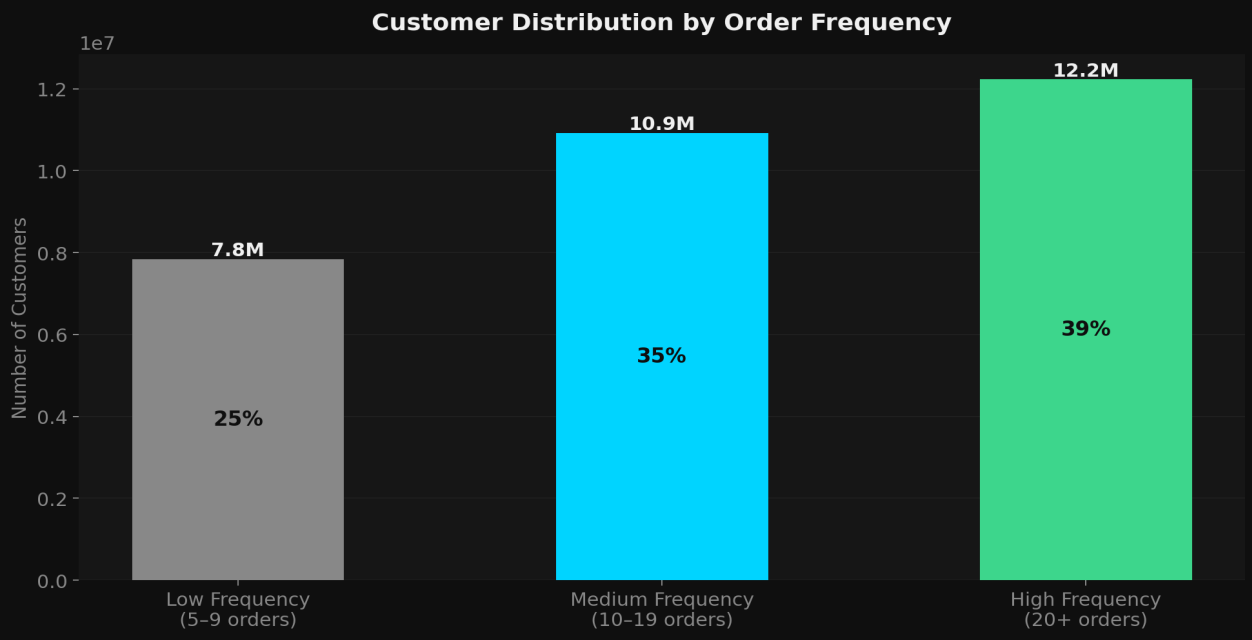
Building Customer Segments

The final profiling step combined all segmentation variables into a unified customer profile string. Low-activity customers (fewer than 5 orders) were excluded first to focus the analysis on revenue-relevant behavior — removing 1.44 million rows.

Combined Profile Format

Each customer was assigned a profile of the form:
Age Group | Income Group | Family Status | Order Frequency

Example: Adult | Middle income | Small family | High frequency



High-frequency customers (20+ orders) are the largest group — repeat buyers drive Instacart revenue.

Figure 8. Customer distribution by order frequency. High-frequency customers (20+ orders) are the largest group — repeat buyers are the backbone of Instacart revenue.

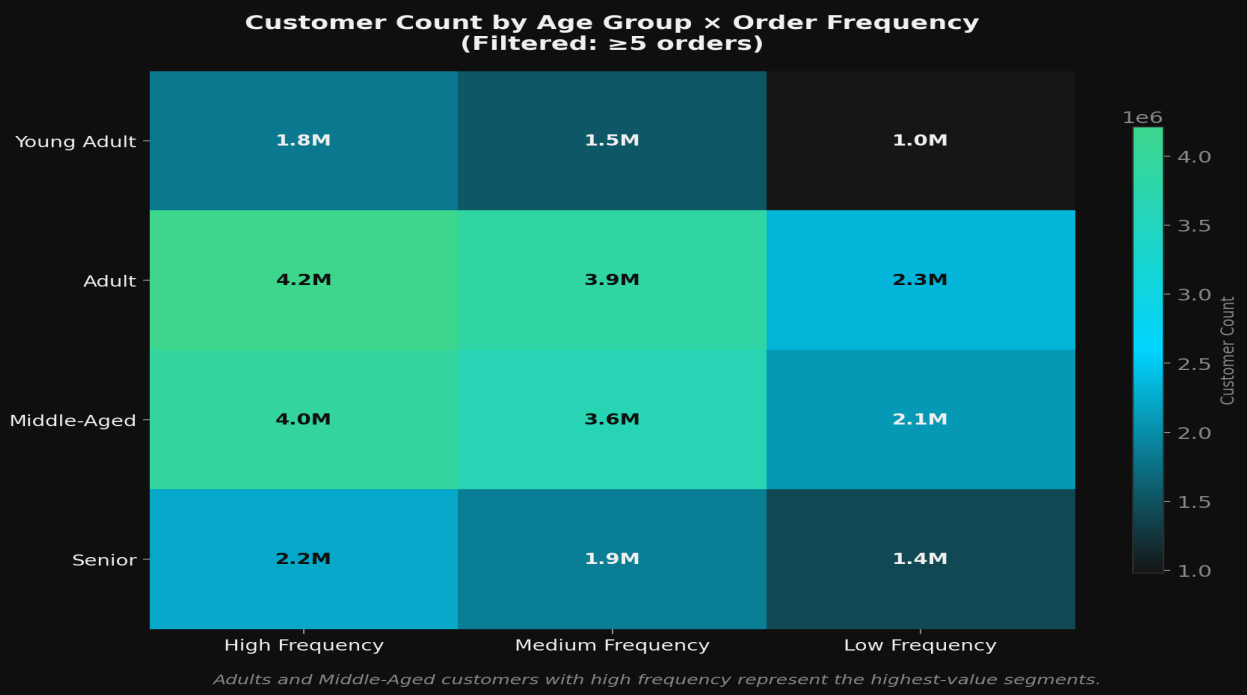


Figure 9. Customer count by age group × order frequency. Adults and Middle-Aged customers with high frequency represent the highest-value, most engaged segments.

KEY INSIGHT

Excluding low-activity customers removed 1.44 million rows (4.4% of data) but significantly improved signal quality. One-time or near-inactive users behave fundamentally differently and would skew segmentation results if included.

07 KEY FINDINGS

Business Insights & Recommendations

n 9 AM and 4 PM.	Instacart is primarily a daytime shopping platform. Late-night o logistics staffing should be concentrated in the 9 AM–4 PM around 10:00–11:00 AM.
le across all hours of the day.	Average basket price does not vary meaningfully by time of d incentives are unlikely to change spending behavior — prom hours.
enerate the most order volume.	While Loyal customers have the highest per-order value, Regu loyalty conversion strategy — moving Regular to Loyal — offers
e dominant customer segment.	This age group represents the largest share of customers anc and marketing campaigns should be optimized for family-stag staples, fresh produce, and meal planning.
mers are the core demographic.	The majority of customers fall in the $40k–$80k income range. target the High income segment specifically, while core catalog Middle income households.
omers are the largest single group.	Customers with 20+ orders outnumber both Medium and Lo strategies (subscriptions, loyalty rewards) have a larger a campaigns, making them higher ROI.

Limitations

- Customer data is synthetic (CareerFoundry dataset) — findings are directional, not production-validated
- No product category-level analysis — department preferences per segment were not fully explored
- Regional findings limited by U.S. customer base — international expansion not addressed
- No time-series component — analysis is a snapshot; seasonal trends are not captured
- Spending flag uses item price as proxy — total basket value per order would be more accurate

APPENDIX

Technical Reference

Dependencies

Library	Purpose
pandas	Data import, wrangling, grouping, merging, export
numpy	Numeric operations, array handling
matplotlib	Bar charts, line charts, scatter plots
seaborn	Styled statistical visualizations
os	File path management across notebooks
pickle / pkl	Efficient storage of large merged dataframes between notebooks

Dataset Summary

Metric	Value
Total order rows (ords_prods_merge)	**~32.4 million**
Total customers	**~206,000**
Orders after low-activity filter	**~31.0 million**
Products	**49,688**
Departments	**21**
U.S. states covered	**50**
U.S. Census regions	**4 (Northeast, Midwest, South, West)**
Customer profile combinations	**Age(4) × Income(3) × Family(3) × Frequency(3) = 108 max**

GitHub & Portfolio

Source code (all notebooks):
github.com/ageelalramadhan/Project-4-Instacart-

Portfolio case study:
ageelalramadhan.github.io/instacart-case-study.html